

Name-Surname: ID:

01286326 THEORY OF COMPUTATION
ASSIGNMENT 2

PROBLEM 1

Suppose M is a finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0,1\}$
- δ is described as

	0	1
q_0	q_1	q_3
q_1	q_2	q_1
q_2	q_4	q_1
q_3	q_3	q_4
q_4	q_0	q_2

- q_0 is the initial state,
- $F = \{q_1, q_4\}$

Show how to construct a regular expression α equivalent to this automaton, i.e. $L(\alpha) = L(M)$.

PROBLEM 2

Let β be the following regular expression over the alphabet $\Sigma = \{0,1\}$:

$$\beta = (01 \cup 011)^+(10 \cup 110)^*$$

Show how to construct a nondeterministic finite automaton N equivalent to this regular expression, i.e. $L(N) = L(\beta)$.

PROBLEM 3

Let R be the following language over the alphabet $\Sigma = \{0,1\}$:

$$R = L((0 \cup 10)^*) - L((\Sigma\Sigma)^*)$$

Find a regular expression φ such that $L(\varphi) = R$.

PROBLEM 4

For each language below, prove or disprove whether it is regular. Here, $\Sigma = \{0,1\}$.

- (1) $L_1 = \{w \in \Sigma^* \mid w \text{ is a palindrome}\}$. Note that a *palindrome* is a string that reads the same forward and backward, such as 10101 and 0110.
- (2) $L_2 = \{0^m 1^n \mid m \geq n \geq 1 \text{ and } m \text{ is divisible by } n\}$.
- (3) $L_3 = \{ww \mid w \in \Sigma^*\} \cap \{w \in \Sigma^* \mid w \text{ contains no consecutive 0s and no consecutive 1s}\}$.